Fed-ZoomSIB with a server: what is (and is not) new

Proofs for claims 1.2, 1.3 and 2.1 of `docs/claims-evidence.md`

CS6007 project notes

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Abstract

We make precise the statement "with a server, N agents for T rounds are a single agent for NT rounds". Theorem 3 shows that no N-agent algorithm, whatever it communicates, can beat the single-agent minimax rate at horizon NT. Theorem 9 shows that Fed-ZoomSIB with exact sufficient-statistic pooling and event-triggered sync *attains* that rate. Theorem 12 bounds its communication by $O(N_{\mathrm{b}} \log_{1+\gamma}(NT))$ sync rounds, and Proposition 15 shows that Phase 1 is an $O(d/(NT)^{1/3})$ fraction of all traffic. Section 5 sketches the serverless (relay) version, whose bound carries an additive $O(NDN_{\mathrm{b}})$ term, and Section 6 states the open regret–communication question these results leave. Imported results from Dey, Bhore & Ghosh (2026), hereafter *DBG*, are marked as such.

1 Setting

N agents, T rounds. At round t agent i observes K contexts $x_{i,t,1}, \dots, x_{i,t,K}$ drawn i.i.d. from $\mathcal{D}$ on $\mathbb{R}^d$, independently across agents and rounds, pulls $a_{i,t}$ and observes $y_{i,t} = f(\langle x_{i,t,a_{i,t}}, \theta^* \rangle) + \eta_{i,t}$. Both θ^* (with $\|\theta^*\|_1 = 1$) and f are unknown and shared by all agents. Network regret is

$$R_T(N) = \sum_{i=1}^{N} \sum_{t=1}^{T} \Big[\max_a f(\langle x_{i,t,a}, \theta^* \rangle) - f(\langle x_{i,t,a_{i,t}}, \theta^* \rangle) \Big].$$

Assumption 1 (DBG Assumptions 1–3)**.** (i) $\eta_{i,t}$ is independent of the contexts and conditionally σ-sub-Gaussian given the past. (ii) $\mathbb{E}[S_j(X)^2] \le M$ for the score $S = -\nabla \log p$. (iii) Each coordinate of X is κ-sub-Gaussian; $|f| \le L_f$ and $|f'| \le L_{f'}$ on $|z| \le X_{\max}$, where $X_{\max} = \kappa\sqrt{2\log(2dNTK/\delta)}$ bounds $\|x\|_\infty$ for all NTK contexts with probability $\ge 1 - \delta$. (iv) $\mu^* = \mathbb{E}[f'(\langle X, \theta^* \rangle)] > 0$.

Lemma 1 (Imported: DBG Lemma B.1 with Lemma 2.1)**.** *Let $\hat\theta$ be the truncated Stein estimator $\frac{1}{n}\sum_{s=1}^{n} \varphi_\tau(y_s S(x_s))$ on n i.i.d. pairs with $\tau = \sqrt{3(\sigma^2 + L_f^2) M n / \log(2d/\delta)}$, and $\hat\theta_0 = \hat\theta / \|\hat\theta\|_1$. If n is large enough that $C_\theta d\sqrt{\log(2d/\delta)/n} \le \mu^*/2$, then with probability $\ge 1 - \delta$*

$$\|\hat\theta_0 - \theta^*\|_1 \le \varepsilon_0(n) := \frac{4 C_\theta d}{\mu^*} \sqrt{\frac{\log(2d/\delta)}{n}}, \qquad C_\theta = \tfrac{4\sqrt{3}}{3} + 2\sqrt{2}\sqrt{M(\sigma^2 + L_f^2)}.$$

Lemma 2 (Imported: DBG Theorem 5.1)**.** *There are $c > 0$ and n_0 such that for every horizon $n \ge n_0$ and every single-agent policy π, $\sup \mathbb{E}[R_n(\pi)] \ge c\, n^{2/3}$, the supremum over instances satisfying the regularity assumptions (with $d = 1$, Gaussian contexts and $K = \lceil 4N' \log n / p_{\min} \rceil$ arms for an N'-bin construction).*

2 Lower bound: a network is never better than one agent with all the data

Theorem 3 (Lower bound by simulation)**.** *Let $\mathcal{A}$ be any N-agent algorithm, with arbitrary communication, including every agent seeing every other agent's data after every round. Then for $NT \geq n_0$,*

$$\sup_{instances} \mathbb{E}[R_T(N;\mathcal{A})] \;\geq\; c\,(NT)^{2/3}.$$

The same holds for every serverless algorithm, since it is a special case.

Proof. We build a single-agent policy π with horizon $n = NT$ whose regret has exactly the law of $\mathcal{A}$'s network regret; Lemma 2 then applies to π.

Index the single agent's rounds by $u = (t-1)N + i$, so that round u plays the role of "agent i at round t". Policy π keeps a copy of $\mathcal{A}$'s full internal state (all agents, the server, and their random seeds, which π draws itself). At round $u = (t, i)$:

1. π observes its context set X_u and passes it to the simulated agent i, which selects a_u using its state *as of the end of round* $t-1$, exactly as in $\mathcal{A}$.
2. π plays a_u and receives y_u, but *withholds* it from the simulation.
3. When $i = N$, all N outcomes of round t are available. π feeds them to the simulated agents' updates and runs one round of $\mathcal{A}$'s communication.

This is a valid policy: each decision uses only information from strictly earlier rounds of π.

The context sets X_u are i.i.d. from $\mathcal{D}^K$, and the noise is independent of everything before it, in both systems. So by induction over t, the joint law of (contexts, actions, rewards) of round t is the same under π and under $\mathcal{A}$, given the same law up to round $t-1$. In particular $R_{NT}(\pi) \overset{d}{=} R_T(N;\mathcal{A})$ on every instance. Hence $\sup \mathbb{E}R_T(N;\mathcal{A}) = \sup \mathbb{E}R_{NT}(\pi) \geq c(NT)^{2/3}$. □

Remark 4. The argument uses only that agents are statistically identical (shared $\theta^*, f, \mathcal{D}$) and act in parallel. It fails under concept shift (different θ_i^*), which is where collaboration stops being "one agent with more data".

3 Upper bound: Fed-ZoomSIB-SS attains the single-agent rate at NT

Algorithm (as analysed).

- *Phase 1* (rounds $1..T_0$): every agent pulls uniformly at random and stores $(S(x), y)$. At round T_0 each agent uploads $b_i = \sum_{t \leq T_0} \varphi_\tau(y_{i,t} S(x_{i,t}))$ and $n_i = T_0$, with τ set for the pooled size $n = NT_0$. The server broadcasts $\hat{\theta}_0 = \sum_i b_i / \|\sum_i b_i\|_1$. (Code: `phase1="exact"`, `freeze="fixed"`.)
- *Grid*: window $W = X_{\max}$; bins $I_j = (-W + (j-1)\Delta,\, -W + j\Delta]$ for $j \leq N_{\mathrm{b}} = \lceil 2W/\Delta \rceil$. An arm x belongs to bin $b(x) \ni \langle x, \hat{\theta}_0 \rangle$.
- *Phase 2*: the server holds the synced table (G_j, G_j^S). Agent i holds unsynced deltas $(\delta n_{i,j}, \delta S_{i,j})$ and acts on $n_{i,j} = G_j + \delta n_{i,j}$, $S_{i,j} = G_j^S + \delta S_{i,j}$. Among the bins its arms fall into, it picks $\arg\max_j \, S_{i,j}/n_{i,j} + w(n_{i,j})$ (with $w(0) = \infty$) and any arm in that bin.
- *Trigger*: at the end of a round, if some agent has $\delta n_{l,j} > \gamma \max(G_j, 1)$, all agents upload their deltas and the server merges and broadcasts them. (Code: `phase2="event"`.)

The confidence width is $w(n) = c_\delta \sqrt{2/n}$ with $c_\delta = \sigma\sqrt{2\log(4NN_{\mathrm{b}}(NT)^3/\delta)}$. (The code uses the smaller $\sigma\sqrt{2\log(2N_{\mathrm{b}}NT/\delta)}$; the difference is a constant and a log factor.)

Lemma 5 (Phase 1 is exactly centralised)**.** $\sum_i b_i/(NT_0)$ *is the truncated Stein estimator on the pooled* $n = NT_0$ *samples. These are i.i.d. with the law of single-agent Phase-1 samples. Hence, on an event* E_θ *of probability* $\geq 1-\delta$, $\|\hat{\theta}_0 - \theta^*\|_1 \leq \varepsilon_0(NT_0)$.

Proof. Truncation is applied per summand with the same τ, so the sum of agents' sums is the sum over the pooled sample. Uniform pulls make each (x, y) an independent draw of the context distribution and the reward model. Apply Lemma 1. □

Good events. E_x: $\|x\|_\infty \leq X_{\max}$ for all contexts. E_θ: as in Lemma 5. E_η: the noise concentration defined in Lemma 6. On $E_x \cap E_\theta$,

$$|\langle x, \hat{\theta}_0\rangle - \langle x, \theta^*\rangle| \leq \|x\|_\infty \|\hat{\theta}_0 - \theta^*\|_1 \leq X_{\max}\varepsilon_0 =: \varepsilon_{\mathrm{p}}, \tag{1}$$

and every arm lies in the window, since $|\langle x, \hat{\theta}_0\rangle| \leq \|x\|_\infty \|\hat{\theta}_0\|_1 \leq W$. Let $\tilde{I}_j = I_j + [-\varepsilon_{\mathrm{p}}, \varepsilon_{\mathrm{p}}]$ and $m_j = \sup_{z\in\tilde{I}_j} f(z)$. By (1) every pull into bin j has true index in $\tilde{I}_j$, so its mean $\mu_s = f(z_s)$ lies in $[m_j - \omega, m_j]$, where $\omega = L_{f'}(\Delta + 2\varepsilon_{\mathrm{p}})$.

Lemma 6 (Confidence for stale, agent-specific tables)**.** *Order all Phase-2 pulls into bin* j *lexicographically by (round, agent). With probability* $\geq 1-\delta$ *(event* E_η*), for every agent* i*, bin* j *and round* t*,*

$$\left|\tfrac{S_{i,j}}{n_{i,j}} - \tfrac{1}{n_{i,j}}\textstyle\sum_{s\in seen}\mu_s\right| \leq w(n_{i,j}).$$

Proof. The pulls agent i sees in bin j are a *prefix* of bin j's sequence (everything up to the last sync) together with a *contiguous segment* of agent i's own bin-j pulls (since that sync). The noise along the lexicographic order is a martingale difference sequence: the bin chosen at (t, i) depends only on rewards of earlier rounds. Azuma–Hoeffding with a union bound over all prefixes ($\leq NT$ per bin) and all segments ($\leq (NT)^2$ per agent and bin) bounds each part's noise sum by $c_\delta\sqrt{\text{length}}$. Then $\sqrt{p} + \sqrt{q} \leq \sqrt{2(p+q)}$ completes the proof. □

Lemma 7 (Per-pull regret)**.** *On* $E = E_x \cap E_\theta \cap E_\eta$*, if agent* i *pulls from bin* J *at round* t*, its instant regret is at most* $\min\{2L_f,\ 2w(n_{i,J}) + 2\omega\}$*. This holds for* any *availability pattern.*

Proof. The best arm x^* lies in an available bin j^* with $f(\langle x^*, \theta^*\rangle) \leq m_{j^*}$. By Lemma 6,

$$\mathrm{UCB}_{j^*} \geq (m_{j^*} - \omega) - w + w = m_{j^*} - \omega,$$

or $\mathrm{UCB}_{j^*} = \infty$. Also $\mathrm{UCB}_J \leq m_J + 2w(n_{i,J})$. The pulled arm has mean $\geq m_J - \omega$. Combining with $\mathrm{UCB}_J \geq \mathrm{UCB}_{j^*}$ gives regret

$$\leq (\mathrm{UCB}_J + \omega) - (m_J - \omega) \leq 2w(n_{i,J}) + 2\omega.$$

Rewards are in $[-L_f, L_f]$, which gives the other bound. □

Lemma 8 (Stale counts)**.** *If a pull is the* k*-th pull into bin* J *(lexicographic order), then the count its agent acts on satisfies* $n_{i,J} \geq (k - c_N)/(1 + N\gamma)$*, where* $c_N = N(1+\gamma)$.

Proof. The $k-1$ earlier pulls are of three kinds:

- synced, G_J;
- agent i's own unsynced pulls;
- other agents' pulls agent i has not seen.

The last kind are either unsynced, at most $(N-1)\gamma\max(G_J, 1)$ because no trigger fired at the end of round $t-1$, or from round t itself, at most $N-1$. Since $n_{i,J} \geq G_J$:

- if $G_J \geq 1$, then $k - 1 \leq n_{i,J}(1 + N\gamma) + N$;
- if $G_J = 0$, then $n_{i,J} \geq k - 1 - (N - 1)(1 + \gamma)$.

Both cases imply the claim. □

Theorem 9 (Upper bound)**.** *Under Assumption 1, with $\delta = 1/(NT)$, $\Delta = (NT)^{-1/3}$, $\gamma = \gamma_0/N$, and $NT_0 = \lceil (L_{f'} X_{\max} \frac{4C_\theta d}{\mu^*} \sqrt{\log(2dNT)}\, NT/L_f)^{2/3} \rceil$ (and at least the size Lemma 1 requires),*

$$\mathbb{E}R_T(N) \leq \underbrace{2L_f NT_0}_{\text{Phase 1}} + \underbrace{4\sqrt{2}\, c_\delta \sqrt{(1+\gamma_0) N_{\mathrm{b}} NT}}_{\text{UCB}} + \underbrace{2L_{f'} \Delta NT}_{\text{discretisation}} + \underbrace{4L_{f'} X_{\max} \varepsilon_0(NT_0) NT}_{\text{projection}} + \underbrace{2L_f N_{\mathrm{b}}(c_N + 1)}_{\text{staleness}} + 6L_f,$$

which is $\tilde{O}\big((d\,NT)^{2/3} + N(NT)^{1/3}\big) = \tilde{O}\big(d^{2/3}(NT)^{2/3}\big)$ for $N \leq \sqrt{T}$. This matches Theorem 3 in N and T up to logarithms: the server setting is "one agent with NT pulls", and no faster.

Proof. Phase-1 pulls cost at most $2L_f$ each. On the failure event (probability $\leq 3\delta$) regret is at most $2L_f NT$, giving the $6L_f$ term. On E, sum Lemma 7 over the pulls into bin j, using Lemma 8 for the k-th pull:

$$\sum_{k=1}^{n_j} \min\Big\{2L_f,\ 2c_\delta \sqrt{\tfrac{2(1+N\gamma)}{k - c_N}} + 2\omega\Big\} \leq 2L_f(c_N + 1) + 4\sqrt{2}\, c_\delta \sqrt{(1 + N\gamma) n_j} + 2\omega n_j.$$

Summing over bins, Cauchy–Schwarz gives $\sum_j \sqrt{n_j} \leq \sqrt{N_{\mathrm{b}} NT}$. Expanding ω gives the discretisation and projection terms. With $\Delta = (NT)^{-1/3}$ we have $N_{\mathrm{b}} \leq 2X_{\max}(NT)^{1/3} + 1$, so the UCB and discretisation terms are $\tilde{O}((NT)^{2/3})$. The choice of NT_0 balances $2L_f NT_0$ against the projection term $\propto d\,NT/\sqrt{NT_0}$; both become $\tilde{O}((dNT)^{2/3})$. Finally, $2L_f N_{\mathrm{b}} c_N = \tilde{O}(N(NT)^{1/3}) \leq \tilde{O}((NT)^{2/3})$ when $N \leq \sqrt{T}$. □

Remark 10 (On the d-dependence)**.** DBG state $\tilde{O}(d^2 T^{2/3})$ for one agent. The cruder per-pull argument above (Lemma 7, projection error charged to every pull) gives a leading term $(dNT)^{2/3}$, plus a lower-order d^2/μ^{*2} burn-in from Lemma 1's sample-size condition. This difference in d has not been cross-checked against DBG's proof and should be verified before it is claimed. The N and T dependence, which is what claim 1.2 concerns, does not rely on it.

Remark 11 (Constant γ)**.** The code and experiments use a constant $\gamma = 0.5$. Then $\sqrt{1 + N\gamma}$ inflates the UCB term by up to $\sqrt{N}$ in the *bound*. E22 finds no such loss in practice: the event-triggered run matches per-round sync.

4 Communication

Theorem 12 (Sync rounds)**.** *Phase 2 of Fed-ZoomSIB-SS performs at most*

$$\sum_{j \text{ used}} \big(1 + \log_{1+\gamma} n_j\big) \ \leq\ N_{\mathrm{b}} \Big(1 + \log_{1+\gamma} \tfrac{NT}{N_{\mathrm{b}}}\Big)$$

sync rounds, where n_j is the final number of pulls in bin j. Per-round sync uses $T - T_0$.

Proof. Charge each sync to one bin j whose trigger fired, i.e. $\delta n_{l,j} > \gamma \max(G_j, 1)$ for some l. The sync adds at least $\delta n_{l,j}$ to G_j:

- if $G_j = 0$, it becomes ≥ 1;
- otherwise G_j grows by a factor of at least $1 + \gamma$.

Since $G_j \leq n_j$, bin j is charged at most $1 + \log_{1+\gamma} n_j$ times. Bins never pulled are never charged. Concavity of log and $\sum_j n_j \leq NT$ give the second form. □

Corollary 13 (Scalars)**.** *Each sync sends 3 scalars per (agent, touched bin) up and 3 per merged bin to each of N agents down. Every pull is uploaded once, so uploads total at most $3NT$. Downloads total at most $3NN_{\mathrm{b}}$ per sync, so at most $3NN_{\mathrm{b}}^2(1+\log_{1+\gamma} NT)$ overall, which is $\tilde{O}(N(NT)^{2/3})$ for $\Delta = (NT)^{-1/3}$. Per-round sync downloads up to $3N\min(N, N_{\mathrm{b}})(T - T_0)$.*

Remark 14 (The bound is loose in practice)**.** E22 measures about 70–120 sync rounds at $NT = 10^4$–$4 \cdot 10^4$, against a bound in the thousands. The busiest bin triggers about 3 syncs, against $1 + \log_{1+\gamma}(NT) \approx 25$. The worst case of Theorem 12 assumes every bin is pulled heavily. Sleeping UCB concentrates pulls on a few near-optimal bins, and most bins trigger at most once. A gap-dependent communication bound is open (Section 6).

Proposition 15 (Phase 1 is a vanishing share)**.** *With fixed T_0, Phase 1 sends $N(2d+3)$ scalars: one $(d+1)$-vector up per agent, $\max|z|$ up, and $(\hat{\theta}_0, W)$ down. Suppose $\gamma < 1$, and let $N_{\mathrm{b}}^{\mathrm{used}}$ be the number of bins ever pulled in Phase 2. Then Phase 2 sends at least $3NN_{\mathrm{b}}^{\mathrm{used}}$ scalars, so*

$$\frac{\textit{Phase-1 scalars}}{\textit{all scalars}} \leq \frac{2d+3}{2d+3+3N_{\mathrm{b}}^{\mathrm{used}}}.$$

Proof. With $\gamma < 1$, a first pull into an empty bin makes $\delta n = 1 > \gamma \max(0, 1)$, so the bin is merged at the next sync and broadcast (3 scalars) to all N agents at least once. □

Since $N_{\mathrm{b}}^{\mathrm{used}}$ grows like $(NT)^{1/3}$ until most of the window is visited, the share is $O(d/(NT)^{1/3})$. Phase 1 matters only when d is comparable with $(NT)^{1/3}$. E24 measures under 2% at $d \leq 100$, and compressing Phase 1 saves at most that share.

5 Without a server: the relay version (sketch)

Dec-ZoomSIB-Relay replaces the server as follows:

- spanning-tree aggregation in Phase 1, which is exact, so Lemma 5 holds after 2·depth extra exploration rounds;
- event-triggered *flooding* of each agent's own bin table in Phase 2: records travel one hop per round, and agent l republishes when its unpublished count exceeds γ times its current view.

Theorem 16 (Relay, sketch)**.** *On a connected graph of diameter D, with the parameters of Theorem 9, the relay version satisfies $\mathbb{E}R_T(N) \leq$ (bound of Thm. 9)$+O(L_f\, ND\, N_{\mathrm{b}})$, i.e. $\tilde{O}\big(d^{2/3}(NT)^{2/3} + ND(NT)^{1/3}\big)$.*

Proof sketch. Lemma 8 gains one more kind of unseen pull: records published by agent l fewer than $\mathrm{dist}(i, l) \leq D$ rounds ago. Each agent makes at most one pull per round, so these number at most $(N-1)D$ per bin. Then $n_{i,J} \geq (k - c_N - ND)/(1 + N\gamma)$, and the first $c_N + ND$ pulls of each bin are charged $2L_f$. Lemma 6 needs the union bound to run over each agent's *view*, which is a union of N per-agent prefixes; this costs a factor $\sqrt{N}$ inside the log-width argument. □

Open. Whether the additive NDN_{b} is necessary is unknown. E23 measures an excess that grows sublinearly in D, is almost flat in T, and is about 100× below NDN_{b}. This suggests the bound is loose: plausibly only Phase-1 delays and the first visits to each bin pay the diameter.

6 The regret–communication trade-off (notes and conjecture)

Theorems 9 and 12 give *one* point on the trade-off curve: optimal regret with $O(N_{\mathrm{b}} \log NT)$ sync rounds. The natural question is the whole curve: with a budget of C sync rounds, what is the best achievable regret?

A heuristic lower bound. Between two syncs, agents share no information. Consider a sync-free window of length L that starts after the network has pooled n samples. During it, the N agents behave like N learners with a common warm start, each learning only from its own L pulls. In the single-agent hard instance (Lemma 2), the bins that must be distinguished have gaps of order $(n+L)^{-1/3}$, and a lone learner needs $L^{2/3}$-type regret to resolve them. That suggests

$$R_T(N;C) \;\gtrsim\; \max\Big\{(NT)^{2/3},\; \max_{\text{windows}} N\,L^{2/3}\Big\},$$

where the largest window is at least T/C if syncs are spread evenly. Under this heuristic, evenly spread syncs need $C \gtrsim \sqrt{N}$ to keep the second term below the first. Windows that grow geometrically could need far fewer.

Conjecture 17. For single-index bandits with a server, the optimal regret $\tilde{\Theta}((NT)^{2/3})$ is achievable with polylog(NT) sync rounds independent of N_{b}. For batched Lipschitz bandits, $O(\log\log T)$ batches suffice (Feng, Huang & Wang 2022; to be checked), but that algorithm eliminates arms it chooses to play. Here arms are given by contexts (sleeping bins), which is exactly what our fixed-schedule experiments (E27) find breaks naive batching.

What a paper would need: (a) a batch-aware Phase 2 (e.g. elimination over bins with context-driven availability) with an upper bound in terms of C; (b) a matching lower bound that formalises the window heuristic. Neither is done here.

Where the experiments are

- Theorem 3/9 in practice: E22 (`fig22`). One agent at NT against N agents at T with the same pooled exploration and bin width.
- Theorem 12: E22 part 2 (`fig22b`), sync rounds and per-bin triggers against $\log NT$.
- Proposition 15: E24 (`fig24`).
- Theorem 16: E23 (`fig23`).
- Section 6: E27 (`fig27`).